

Yumi Heo

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Data Scientist

Data scientist working with real-world patient data to develop clinical prediction models. At AITIS, I collaborate with hospitals in Singapore and clinical specialists on an ongoing project to predict healing time for chronic wounds, alongside developing skin cancer classification models and deploying models with software engineers. Previously, at the Office of Health Economics, I built Python and RAG pipelines to automate the collection and summarisation of pharmaceutical regulatory documents.

SKILLS

Languages & Tools: Python, SQL (PostgreSQL), R, Matlab, Git, Docker, FastAPI, Jira, Confluence

ML & CV: PyTorch, TensorFlow/Keras, OpenCV, Torchvision, Pillow, ONNX, Scikit-learn, Pandas, NumPy

LLM & Agentic AI: HuggingFace, LangChain, LangSmith, LangGraph, MCP

Visualisation: Matplotlib, Seaborn, Plotly, Tableau, Folium

Cloud & Big Data: AWS (RDS, EC2, S3), GCP (BigQuery, Compute Engine, Cloud Run, Cloud Storage), PySpark

PROJECTS

- **NoteGuard:** Built a clinical text de-identification pipeline combining rule-based detection and patient-consistent pseudonymisation, enabling privacy-preserving use of free-text clinical data
- **Automated-EMA-NICE-Data-Scraping:** Built an automated Python pipeline using GPT-4o mini to extract and summarise unstructured data from the European Medicines Agency, significantly improving data collection efficiency
- **Inventory Placement Optimiser:** Developed a prototype for an inventory placement optimiser using demand forecasting, multinomial logistic regression and linear programming to minimise fulfilment costs under warehouse capacity constraints

WORK EXPERIENCE

Data Scientist, AITIS, London, UK

Jan 2025 - Present

AITIS develops clinical AI technology and health-monitoring wearables. I develop classification and forecasting models in collaboration with doctors and dermatologists.

- Collaborating with hospitals in Singapore on an ongoing project using real-world patient data to predict time to healing for chronic wounds.
- Reduced clinician wound image annotation workload by 90% by developing a sample prioritisation tool that highlighted the hardest-to-classify images for human review
- Developed skin cancer classification models achieving over 90% sensitivity and specificity by addressing class imbalance using a variety of data augmentation techniques
- Deployed models to production via FastAPI and Docker with unit tests, working with software engineers in an agile team
- Mentored summer interns to support their onboarding and personal development

Data Science Intern, Office of Health Economics, London, UK

Jul 2024 - Oct 2024

The Office of Health Economics is an independent health economics research organisation. I developed an automated data collection tool to support health economists.

- Cut data collection time from 2 days to 10 minutes by building an automated pipeline to collect pharmaceutical regulatory data from the National Institute for Health and Care Excellence and the European Medicines Agency
- Engineered a RAG pipeline with Chroma and LangChain to improve retrieval and summarisation accuracy on medical documents
- Improved summarisation quality through prompt engineering and evaluation by scoring outputs with ROUGE
- Conducted hands-on training sessions allowing non-technical colleagues to operate the workflow independently

Acquisition & Distribution Assistant Manager, Danal Entertainment, South Korea

Apr 2019 - Oct 2022

Danal Entertainment distributes films, TV series and music in South Korea. I supported commercial decision-making and managed the distribution of 100+ films and TV series across Netflix, Google Play and Naver.

- Developed statistical forecasting models to predict film profitability across a 10,000+ film catalogue; achieved a 45–65% recoupment of licence fees at or before release
- Negotiated licensing agreements with 50+ international film providers, leveraging forecast insights to evaluate commercial opportunities and secure an exclusive distribution partnership with Coupang Play
- Prepared monthly and annual sales-performance reporting, translating complex data into acquisition and pricing recommendations for senior management

EDUCATION

MSc Data Science (Distinction), City St George's, University of London, UK

Sep 2023 - Jan 2025

- Dissertation (72%): Developed an interpretable ensemble anomaly detection framework for industrial control systems using DBSCAN labelling, LSTM-GRU stacking and SHAP/LIME explainability
- Modules: Natural Language Processing (88.5%), Machine Learning (81%), Big Data (78%), Neural Computing (76%), Visual Analytics (75%)

BA Economics and English Language & Literature (2:1), Dongguk University, Seoul, South Korea

Mar 2015 - Feb 2018